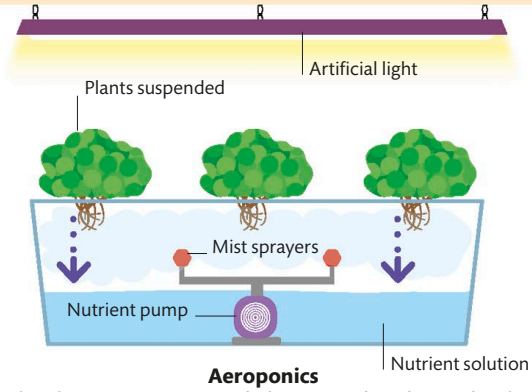
# Farming without soil

**As demand for food increases, farmers are devising more efficient methods for growing crops. Farming without soil allows farmers to grow plants almost anywhere, carefully controlling growing conditions and minimizing environmental impact.**

**A HYDROPONIC FARM USES JUST 10 PERCENT OF THE WATER USED ON A CONVENTIONAL FARM**

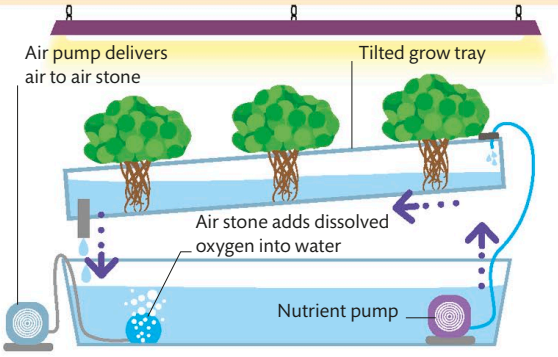
## Hydroponics

In a hydroponic system, crops are grown without soil and are fed by nutrients dissolved in water that is usually delivered by a pump. Nutrient levels can be tailored to the plant type, and lighting, ventilation, humidity, and temperature are easy to control. There are several different kinds of hydroponic systems.



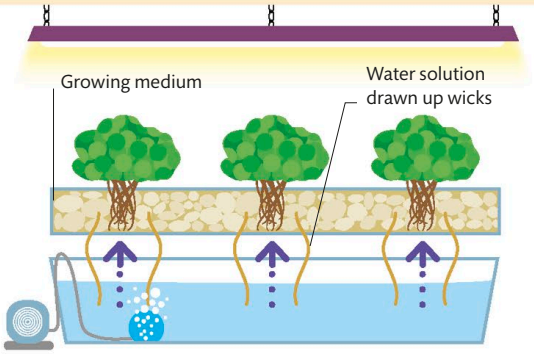

**Aeroponics**

The plant roots are suspended over a tank and misted with nutrient solution delivered by a nutrient pump. They are misted every few minutes to prevent the roots from drying out.



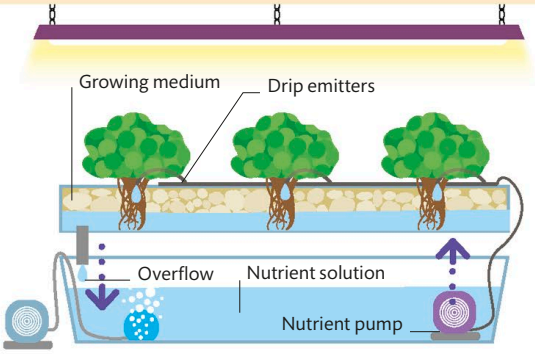

**Nutrient film technique**

The nutrient solution is pumped into a grow tray and continually flows over the tips of the roots. The tray is set at an angle so that the water flows back into the tank under gravity.




**Wick system**

Plants are grown in a medium of perlite, coir, or vermiculite. The nutrients are drawn up from the tank to the growing medium by the capillary action of absorbent wicks.




**Drip system**

Nutrients are dripped onto the growing medium around each plant at regular intervals. Excess solution runs off and can be recycled back into the system.